

AE-Multi-WGAN: A Robust Multi-Attack Detection IDS for Imbalanced Datasets

Ishfaq Bashir Sofi*, *Student Member, IEEE*, Anjali Agarwal*, *Member, IEEE*, Kuljeet Kaur^{†‡}, *Member, IEEE*

*Department of Electrical and Computer Engineering, Concordia University, Montreal, QC, Canada

[†]Department of Electrical Engineering, École de technologie supérieure, Montreal, QC, Canada

[‡]Canadian University Dubai, City Walk, Dubai, United Arab Emirates

Email: ishfaq.sofi@mail.concordia.ca, aagarwal@ece.concordia.ca, kuljeet.kaur@etsmtl.ca

Abstract—Intrusion Detection Systems (IDS) are vital for identifying malicious activities in complex network environments. Numerous approaches, from traditional Machine Learning (ML), statistical techniques, and Deep Learning (DL) models, have been developed to detect attacks using IDSs. However, existing approaches suffer from high false positive (FP) rates, lower accuracy, and heavy computational load. Amongst them, Generative Adversarial Networks (GANs) have gained significant interest due to their ability to model complex data distributions and generate realistic synthetic samples. However, GAN-based IDS methods still face key challenges, such as class imbalance, mode collapse, training instability, and difficulty in capturing diverse attack patterns, which limits their generalization and reliability in practice. To address these issues, we propose an Ensembled AutoEncoder Multi-Wasserstein GAN (AE-Multi-WGAN) scheme, integrating multiple generators and discriminators with an AutoEncoder (AE) in an ensemble DL framework to improve attack detection. The AE enables denoising and feature compression, while the WGAN component ensures stable training and high-quality synthetic sample generation. Validated on the NSL-KDD and UNSW-NB15 datasets, the proposed AE-Multi-WGAN demonstrates superior precision, recall, and F1-score compared to baseline methods, effectively enhancing IDS performance, reducing FP, and improving generalization to unseen attacks.

Index Terms—AutoEncoder, Generative Adversarial Network, Intrusion Detection System, Imbalanced Data Set.

I. INTRODUCTION

With the rapid growth of Internet technologies, such as the Internet of Things (IoT), data transmission, communication systems, and cybersecurity face significant challenges. The exponential growth of network activity has led to a surge in cyberattacks, resulting in financial losses, service disruptions, and other adverse impacts on organizations and countries [1]. Thus, the demand for robust network protection has grown as hackers, empowered by rapid technological advancements and easy information exchange, have developed more sophisticated cyberattacks, further intensified by increased internet usage. As a result, Network Intrusion Detection Systems (NIDS) [2] have become critical components in modern cybersecurity to detect and mitigate potential threats. In addition, advances in ML and DL, paired with the explosion of data volumes, have further enhanced the capabilities of anomaly-based NIDS. When trained on sufficient data, DL models often outperform traditional ML algorithms for attack classification and prediction [3]. However, these models face a significant

challenge when dealing with imbalanced data, where the majority class dominates the minority classes. This imbalance can further lead to poor detection of rare attacks, as the model tends to favour the majority class during training. This issue is evident in widely used IDS datasets such as NSL-KDD [4] and UNSW-NB15 [5]. This class imbalance causes learning algorithms to bias toward majority classes, reducing their ability to detect rare or emerging threats.

To address this, recent research has explored various ML and DL techniques that aim to improve detection accuracy, especially for underrepresented attack types. Zhang *et al.* [6] reviewed deep models such as Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), acknowledging their power in pattern recognition but also their tendency to overlook minority classes due to class imbalance. Selvakumar *et al.* [7] introduced an ensemble framework combining Feature-Augmented CNN with a Deep Autoencoder, which enhanced minority class detection; however, it introduced additional high computational costs. Similarly, Zheng *et al.* [8] developed SF-IDS, a semi-supervised approach using hybrid loss and self-training mechanisms that improved fine-grained detection but suffered from noisy pseudo-labeling. In another attempt, Nawaz *et al.* [9] attempted to mitigate imbalance by integrating Synthetic Minority Over-sampling Technique (SMOTE) with Long Short-Term Memory (LSTM) [10] and categorical focal loss; however, synthetic oversampling techniques like SMOTE risk overfitting by generating less diverse or redundant samples. In contrast, Wang *et al.* [11] introduced S2CGAN-IDS, a GAN-based approach tailored for IoT environments, which effectively synthesized minority class data, though it faced GAN-related issues, including unstable training and inconsistent sample quality. Alternatively, Kaushik *et al.* [12] proposed a lightweight IDS using statistical feature selection to improve efficiency, but this approach lacked the sophistication to model complex and nonlinear traffic patterns typical in modern attacks.

Expanding on generative approaches, GANs have shown significant promise in tackling class imbalance by synthesizing realistic minority attack samples. For instance, the early model GAN-IDS [13] applied a basic GAN to generate attack data but struggled with mode collapse, training instability, and poor minority-class detection. Building on this, WGAN-IDS [14] addressed these issues using Wasserstein loss for

smoother optimization, yet it lacked conditional mechanisms for targeting specific attack types. To improve this further, AE-WGAN [15] incorporated an AE into the GAN framework, which enhanced training stability and mode coverage, though the combined model resulted in increased training complexity and computational overhead. Taking a different route, Multi-GAN IDS [16] introduced multiple generator-discriminator pairs, each dedicated to a different class. While this enhanced minority class representation, it introduced higher computational requirements and risked imbalance among the generators. To address training stability, SNGAN [17] employed spectral normalization to improve generalization but did not enforce class-wise separation, resulting in potential feature overlap. Further, MAGENTO [18] adopted a multi-objective approach optimizing both detection accuracy and data quality, but fine-tuning multiple objectives simultaneously made training sensitive to hyperparameter settings. Similarly, TACGAN-IDS [19] introduced temporal conditioning for sequential attack modelling, yet required extensive temporal labelling and suffered from inefficiency in training. Most recently, TMG-IDS [20] integrated transformer models with multi-generator GANs to enhance class-wise diversity and attention-based learning. However, despite notable advancements, existing GAN-based IDS models continue to face challenges in balancing performance, scalability, and real-world applicability.

A. Motivation

IDSs are essential for detecting malicious activities in network environments but face challenges in identifying rare or novel attacks due to class imbalance, where minority attack types are underrepresented. This imbalance leads to poor detection performance, especially for minority classes, when using traditional ML/DL models. Generative approaches like GANs offer promise for data augmentation but often struggle with training instability and mode collapse. To overcome this, the proposed approach introduces an AE-Multi-WGAN architecture, where the AE compresses and denoises high-dimensional input into meaningful latent features, and the Multi-WGAN generates diverse and high-quality synthetic samples for underrepresented classes. This integrated framework not only addresses class imbalance but also ensures stable adversarial training, thereby improving detection accuracy and enhancing the overall reliability of IDS models.

B. Contributions

The key contributions of the work are outlined below:

- 1) Propose a hybrid AE-Multi-WGAN that combines AE-driven feature learning with ensembled WGANs to generate realistic synthetic attacks and detect rare intrusions.
- 2) Tackle class imbalance by producing high-quality samples for minority classes while mitigating mode collapse and training instability.
- 3) Achieve improved IDS performance with reduced False Positives (FP), higher accuracy, and better generalization to unseen threats.

C. Organization

The rest of the manuscript is organized as follows. Section II presents the system model for the proposed scheme, followed by a detailed evaluation on different datasets in Section III. Finally, the work is concluded in Section IV.

II. SYSTEM MODEL

This section presents the architecture of the proposed AE-Multi-WGAN-based IDS, illustrated in Fig. 1. It includes four modules: i) Data Preprocessing, ii) AE-based Oversampling, iii) Classification of Minority Classes, and iv) Multi-WGAN. Together, these components are designed to detect anomalies, handle class imbalance, stabilize training, and mitigate mode collapse in network traffic analysis. These modules are described in the following.

Data preprocessing is fundamental in developing an effective IDS, especially when handling heterogeneous and imbalanced network traffic. The raw dataset comprises numeric and categorical features, along with challenges like noise, missing values, and skewed class distributions [15]. The data goes through a structured sequence of preprocessing stages to ensure model-ready input. The process begins with data cleaning, where missing values, duplicates, format inconsistencies, and outliers are addressed to ensure consistency and quality. This step prepares the dataset for further transformation and analysis. Redundant features are removed by discarding those with fixed values (constant features) or low variance (quasi-constant features). Categorical features are then converted into numeric form using an ordinal encoder [21]. After encoding, numeric features are scaled using Min-Max normalization, which brings values into a uniform range. This normalization ensures consistency across features and accelerates learning by reducing variability in feature magnitudes. Once encoding and scaling are complete, the dataset is fully numeric and suitable for ML/DL models. Feature selection follows, using Analysis of Variance (ANOVA) [22], which identifies statistically significant features by evaluating variance relative to the target variable. This reduces dimensionality and enhances both model efficiency and detection accuracy. The final output is a clean, normalized, and feature-optimized dataset, effectively prepared for model training and synthetic data generation.

A. Module 2: AutoEncoder-based Oversampling

In the proposed scheme, the AE plays a central role in enhancing the quality of adversarial attacks and improving anomaly detection performance, particularly under imbalanced data conditions. The AE contributes through two main functions:

- 1) **Denoising:** The encoder filters noise by mapping the input to a cleaner latent representation, improving the stability and quality of synthetic samples generated by the GANs.
- 2) **Feature Compression:** By reducing input dimensionality, the AE enables the GANs to focus on critical patterns, especially for rare attack instances.

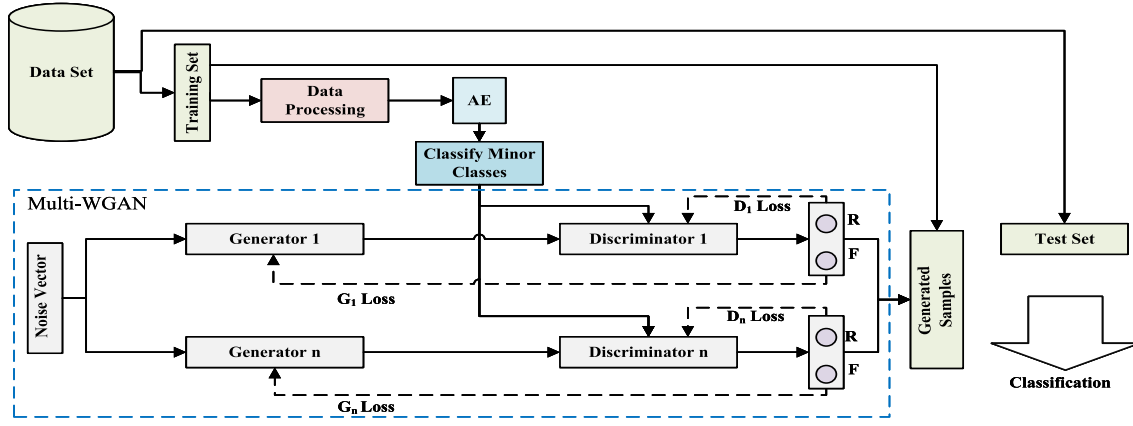


Fig. 1: An illustration of the proposed system model.

The AE is integrated into the Multi-WGAN to enhance the quality of generated attack samples by minimizing noise within the scheme. Architecturally, the AE consists of two main components: the encoder E , which maps the input x to a latent representation h using the function:

$$h = f(Wx + b) \quad (1)$$

here, f is the activation function, W is the weight matrix of the encoder, x is the input vector, and b is the bias term. The decoder D reconstructs the original input x' from h , as shown below.

$$x' = f'(W'h + b') \quad (2)$$

here, W' is the weight matrix of the decoder, f' is the activation function used in the decoder (often the same as f), and b' is the decoder bias term.

This reconstruction helps reduce classification error in the discriminator, improving adversarial training and updating the generator's loss during backpropagation. The non-linear mapping in E allows the AE to learn abstract attack features. Figure 2 shows the AE architecture: the encoder has three dense layers (24, 20, and 16 neurons), while the decoder uses the reverse order (16, 20, and 24 neurons). All layers use ReLU activation. AE is trained with the Adam optimizer with the learning rate of 1×10^{-4} and batch size of 64, over 100 epochs. The training minimizes reconstruction error using Mean Square Error (MSE) loss, wherein the symbols x_i , $\hat{x}_i$, and n refer to the original input, the reconstructed output, and the total number of input samples, respectively.

$$\mathcal{L}_{AE} = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{x}_i)^2 \quad (3)$$

This enables compact, denoised representations and supports realistic, diverse synthetic sample generation by the WGAN.

B. Module 3: Classify Minority Classes

In this module, a minority class classification model is leveraged to identify and classify underrepresented classes. This module addresses class imbalance challenges, improving the detection and classification of minority classes based on

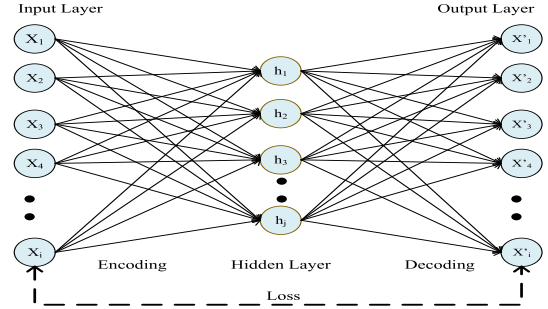


Fig. 2: An illustration of the employed AE structure.

label distribution. Classes with significantly fewer instances are flagged as minority classes, particularly those representing less than 10% of the training samples. In the NSL-KDD dataset, five classes exist, *i.e.*, Normal, Probe, DoS, U2R, and L2R. Among them, Probe, U2R, and L2R represent less than 10% of the training data and are classified as minority classes. In the UNSW-NB15 dataset, which includes nine attack categories, the last six, Shellcode, Worms, Backdoors, Analysis, Generic, and Exploits, fall below the 10% threshold and are also considered minority classes due to their limited representation.

C. Module 4: Multi-Wasserstein GAN

To address class imbalance and capture diverse attack patterns, the proposed scheme employs multiple generators G and discriminators D , each dedicated to different classes within the data distribution. This ensemble architecture enhances diversity, improves minority class coverage, and reduces mode collapse. By dividing the learning task among multiple G - D pairs, Multi-WGAN captures complex data patterns while maintaining training stability. It retains the Wasserstein loss, replacing the Jensen-Shannon (JS) divergence with the Earth-Mover distance, which provides smoother gradients and mitigates vanishing gradients [14]. This improves robustness by accurately measuring the discrepancy between real and generated data distributions. Each G - D pair in the proposed scheme incorporates CNN-based architectures to

enhance training stability, reduce computation, and improve overall efficiency.

Uniformity in G and D design enhances the model's ability to distinguish between real and synthetic data. AE-Multi-WGAN integrates ensemble learning, noise reduction, and Wasserstein loss to improve adversarial training under noise and instability, ensuring reliable anomaly detection. The training workflow involves an adversarial feedback loop between G -Loss and D -Loss. This interaction helps the G refine its output over iterations, making samples more realistic while the D enhances its ability to differentiate real from fake inputs. As proposed in [14], WGAN replaces JS divergence with the Wasserstein distance (Earth-Mover distance), offering more stable training and addressing mode collapse. The distance reflects the effort required to transform one probability distribution into another, improving alignment between real and generated data. In this setup, each G receives a uniform noise vector from the range $[0, 1]$. Although noise input remains the same, each generator is trained independently with feedback from its assigned discriminator, enabling class-specific learning and diverse sample generation.

The WGAN objective minimizes the Wasserstein-1 (Earth-Mover) distance between the real data distribution and the generated one, offering more stable and meaningful training than traditional GANs:

$$\min_G \max_{D \in \mathcal{D}} \mathbb{E}_{x \sim P_r} [D(x)] - \mathbb{E}_{z \sim P_z} [D(G(z))] \quad (4)$$

Here, the Generator G maps a noise vector $z \sim P_z$ (typically drawn from a standard Gaussian or uniform distribution) to a synthetic data sample $G(z)$. The discriminator D , is trained to assign higher scores to real samples $x \sim P_r$ and lower scores to generated samples $G(z)$, effectively learning to distinguish between the two. The term $\mathbb{E}_{x \sim P_r} [D(x)]$ represents the expected output of the critic on real data, while $\mathbb{E}_{z \sim P_z} [D(G(z))]$ represents the expected output on generated data. The constraint $D \in \mathcal{D}$ ensures that D belongs to the set of 1-Lipschitz functions, which is essential for approximating the Wasserstein distance. This constraint is commonly enforced using weight clipping or a gradient penalty, and it helps stabilize the training process by maintaining well-behaved gradients and reducing the risk of mode collapse.

In Multi-WGAN, where multiple G - D pairs handle diverse class distributions, the loss functions are defined as:

$$\mathcal{L}_D = \mathbb{E}_{x \sim P_r} [D(x)] - \mathbb{E}_{\tilde{x} \sim P_g} [D(\tilde{x})] \quad (5)$$

$$\mathcal{L}_G = -\mathbb{E}_{\tilde{x} \sim P_g} [D(\tilde{x})] \quad (6)$$

Here, $\mathcal{L}_D$ represents the discriminator loss, which maximizes the difference between outputs on real data $x \sim P_r$ and generated data $\tilde{x} = G(z) \sim P_g$. In contrast, $\mathcal{L}_G$ is the generator loss, which minimizes this difference to make generated samples indistinguishable from real ones. The generator and discriminator engage in a minimax game where each improves iteratively, D learns to distinguish, while G

learns to deceive. Over time, the generated distribution P_g approximates the real distribution P_r , leading to improved sample quality, training stability, and diversity.

III. PERFORMANCE EVALUATION AND RESULTS

This section presents the performance evaluation of the proposed scheme, covering the details of the dataset used, baseline approaches, and a discussion of results. Further, to ensure all minority classes are represented, the number of G - D pairs matches the count of minority classes per dataset, three for NSL-KDD and six for UNSW-NB15. The CNN-based G - D architectures remain consistent across all pairs and are configured with shared parameter settings, as detailed in Table I, CNN-based G and D parameters.

TABLE I: CNN-based G and D parameters.

Parameter	Value	Parameter	Value
Input neurons	Same as data dimension	Hidden layers	32, 64, 128, 256
Batch size	64	Loss function	ReLU, Sigmoid
Steps/epoch	[0.01, 0.001]	Learning rate	Wasserstein loss
Epochs	100	Optimizer	Adam

A. Dataset Used

The benchmark datasets NSL-KDD [4] and UNSW-NB15 [5] are used in this paper to evaluate the proposed scheme.

1) *NSL-KDD Dataset*: The NSL-KDD dataset, proposed in [4], is an improved version of KDD CUP'99, addressing redundancy and bias. It contains 1,25,973 training and 22,544 testing samples with 41 features. As shown in Fig. 3, the dataset has five classes and suffers from class imbalance, with DoS and Normal classes dominating Probe, U2R, and R2L classes.

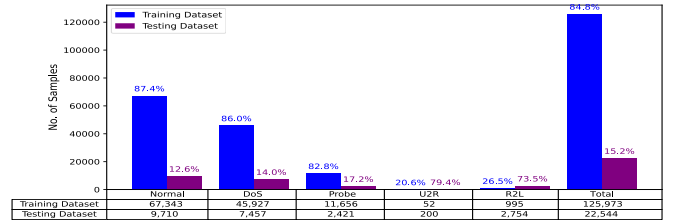


Fig. 3: Distribution of different classes in NSL-KDD.

2) *UNSW-NB15 Dataset*: The UNSW-NB15 dataset, created by the Australian Centre for Cyber Security (ACCS) [5], addresses limitations in older datasets like KDD CUP'99. It offers realistic network traffic and diverse attack types, comprising 1,75,341 training and 82,332 testing samples, each with 49 features. As shown in Fig. 4, the dataset includes nine attack categories, with six considered minority classes due to their limited sample representation.

B. Baseline Approaches

To evaluate the proposed scheme, several baseline models are used for comparison. For NSL-KDD, AE-WGAN [3]

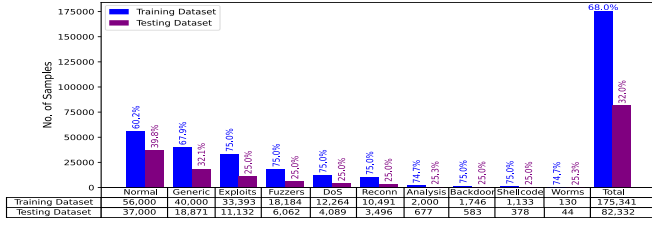


Fig. 4: Distrubution of different classes in (UNSW-NB15).

serves as the baseline, to address data imbalance. For UNSW-NB15, the baseline includes SNGAN [17], representing over-sampling. Additionally, recent GAN-based IDS models, namely, MAGENTO [18] TMG-IDS [20] and TACGAN-IDS [19], are also included for comparative evaluation.

TABLE II: An illustration of duplicated generated samples.

NSL-KDD Dataset		
Minority classes	Generated samples	Percentage
Probe	8519	6.33%
U2R	22169	14.96%
R2L	21061	14.32%

UNSW-NB15 Dataset		
Minority classes	Generated samples	Percentage
DoS	5856	3.23%
Reconn	7826	4.27%
Analysis	17260	8.96%
Blackdoor	17543	9.10%
Shellcode	18223	9.41%
Worms	19338	9.93%

C. Discussion of Results

For extensive evaluations, we present the results using the following three categories:

1) *Performance comparison using different ML/DL classifiers*: To begin, we analyze the quality of the generated synthetic samples by evaluating the extent of duplication within the dataset. Table II reports the percentage of duplicated samples for both datasets. The presence of these duplicated samples indicates that the proposed scheme has generated synthetic data samples to address the class imbalance. Following this, multiple attacks are classified using different pre-existing ML and DL classifiers. The goal is to assess each model's ability to detect and classify anomalies while minimizing FP and FN. Among ML classifiers, XGBoost performs best on both NSL-KDD and UNSW-NB15 datasets, as illustrated in Fig. 5 and Fig. 6, respectively. On NSL-KDD, it achieves around 86% accuracy and 87% precision, benefiting from its sequential boosting mechanism, which corrects misclassifications, manages class imbalance, and avoids overfitting. On the UNSW-NB15 dataset, XGBoost reaches over 93% precision and 87% accuracy, showing strong performance on structured, high-dimensional data. Among DL models, CNN achieves the best results on the NSL-KDD dataset, reaching nearly 86% accuracy and recall by capturing spatial hierarchies. Although its precision and F1-score are slightly below XGBoost, CNN outperforms GRU, LSTM, and RNN. On UNSW-NB15 dataset, CNN and RNN both delivers 88% accuracy,

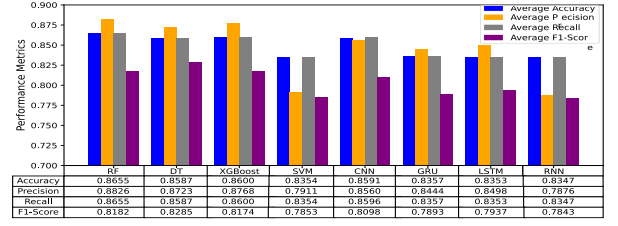


Fig. 5: Performance of proposed scheme (NSL-KDD).

with recall and F1-scores ranging from 87–88%, making them the top DL performers. However, RNN performs poorly on NSL-KDD, with precision and F1-score below 78%, but improves significantly on UNSW-NB15 dataset. In contrast,

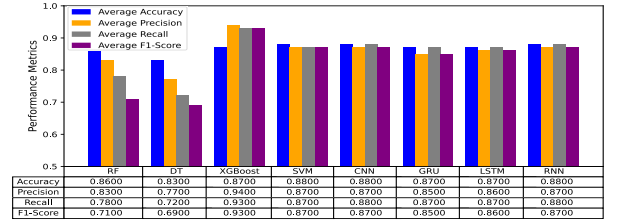


Fig. 6: Performance of proposed scheme (UNSW-NB15).

DT performs weakest on UNSW-NB15, showing the lowest F1-score and recall, which indicates poor generalization and a tendency to overfit on imbalanced data. Overall, XGBoost and CNN demonstrate consistent performance across datasets, while RNN and DT vary depending on dataset characteristics.

TABLE III: Performance comparison between AE-WGAN and Proposed Scheme (NSL-KDD)

Classifier	Scheme	Precision	Recall	F1-score
RF	AE-WGAN	0.73	0.64	0.63
	Proposed	0.83	0.78	0.71
DT	AE-WGAN	0.55	0.56	0.52
	Proposed	0.77	0.72	0.69
CNN	AE-WGAN	0.68	0.65	0.65
	Proposed	0.87	0.88	0.87
GRU	AE-WGAN	0.71	0.69	0.69
	Proposed	0.85	0.87	0.85
LSTM	AE-WGAN	0.75	0.69	0.69
	Proposed	0.86	0.87	0.86
RNN	AE-WGAN	0.70	0.66	0.66
	Proposed	0.87	0.88	0.87

2) *Performance comparison against AE-WGAN on different datasets*: Table III illustrates the performance evaluation of the proposed scheme against AE-WGAN across different metrics using different ML/DL classifiers. Precision improvements are especially notable in DL models, such as CNN, RNN, and LSTM, where it reaches above 86%, exceeding AE-WGAN by more than 10%. Among ML models, DT achieves 77% precision compared to AE-WGAN's 55%, while RF attains 83%, about 10% higher. Recall scores also favor the

proposed scheme, with 78–88% on NSL-KDD, where CNN and RNN achieved highest 88%, outperforming AE-WGAN's more than 15%. These improvements highlight better anomaly detection and handling of class imbalance. F1-scores range between 85–87% in DL classifier, indicate a strong balance between precision and recall, while AE-WGAN remains below 70%, confirming the proposed scheme's higher reliability in detecting minority class attacks.

TABLE IV: Performance comparison of GAN-based IDS methods with proposed scheme (UNSW-NB15)

SNGAN [17]			MAGENTO [18]		
Precision	Recall	F1-score	Precision	Recall	F1-score
0.6424	0.7257	0.6423	0.6594	0.7868	0.7095
TACGAN-IDS [19]			TMG-IDS [20]		
Precision	Recall	F1-score	Precision	Recall	F1-score
0.6197	0.7563	0.6493	0.7162	0.5682	0.7496
AE-Multi-WGAN (ML)			AE-Multi-WGAN (DL)		
Precision	Recall	F1-score	Precision	Recall	F1-score
0.9400	0.9300	0.9300	0.8700	0.8800	0.8700

3) *Performance comparison against different GAN-based IDS*: In this section, we evaluate the performance of the proposed AE-Multi-WGAN against existing state-of-the-art GAN-based IDS models, including SNGAN [17], MAGENTO, [18], TACGAN [19] and TMG-IDS [20]. Table IV presents a detailed comparison. AE-Multi-WGAN consistently outperforms other methods. The ML classifier XGBoost achieves the best results with 94% precision, 93% recall, and 93% F1-score, demonstrating strong attack detection and low FP rates. Among DL classifiers, CNN and RNN perform well (87% precision, 88% recall, 87% F1-score), but slightly lag behind XGBoost, suggesting better ML generalization on synthetic data. In contrast, methods like TMG-IDS, TACGAN-IDS, MAGENTO, and SNGAN show trade-offs. MAGENTO achieves a 70% F1-score, while TMG-IDS reaches 74%. TACGAN and SNGAN boost recall but generate noisy samples, resulting in low precision and an F1-score of 64%. Overall, AE-Multi-WGAN (ML) proves most effective, generating high-quality samples and enhancing the detection of minority class attacks in imbalanced datasets.

IV. CONCLUSION

In this paper, an ensemble-based AE-Multi-WGAN scheme for NIDS is proposed, targeting class imbalance and training instability. By integrating AE with Multi-WGAN in an ensemble DL framework, the scheme generates high-quality synthetic samples for minority classes, improves feature representation, and simultaneously mitigates class imbalance and model collapse. Experiments on NSL-KDD and UNSW-NB15 validate its effectiveness. The proposed scheme achieves notable improvements in precision, recall, and F1-score compared to baseline methods. These results confirm the robustness and reliability of AE-Multi-WGAN in enhancing IDS

performance while effectively addressing major challenges in anomaly-based intrusion detection. Future work will focus on extending the framework to support real-time detection and its deployment in edge and IoT environments.

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